H4: Electric fields in matter

4.1 polarization

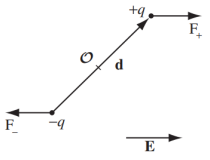
4.1.1 dielectrics

conductor	= contains 'unlimited' supply of charges and conduct electrical charge
dielectric	= all charges are attached to specific atoms or molecules > microscopic displacements aren't as dramatic, but does have a characteristic behaviour

4.1.2 induced dipoles

ionization	place a neutral atom in an electric field > the atom will get polarized, ie: positive and negative side > atom has a dipole moment **p** in the same direction as E, approximately: $$\mathbf{p} = \alpha \mathbf{E}.$$ with α the atomic polarizability
polarizability tensor	When the field is at some angle, we have to split into parallel and perpendicular components: $$\mathbf{p} = \alpha_\perp \mathbf{E}_\perp + \alpha_\parallel \mathbf{E}_\parallel.$$ however, the most general linear relation between **E** and **p** is: $$\left.\begin{aligned} p_x &= \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z \\ p_y &= \alpha_{yx}E_x + \alpha_{yy}E_y + \alpha_{yz}E_z \\ p_z &= \alpha_{zx}E_x + \alpha_{zy}E_y + \alpha_{zz}E_z \end{aligned}\right\}$$ with α_{ij} the polarizability tensor

4.1.3 alignment of polar molecules

force on polar molecules in uniform field	consider a molecule with a permanent dipole moment, ex water > put these molec in a uniform field > the force on the positive end is F_+ = qE and the negative is F_-=-qE > there will be a torque: $$\begin{aligned}\mathbf{N} &= (\mathbf{r}_+ \times \mathbf{F}_+) + (\mathbf{r}_- \times \mathbf{F}_-) \\ &= \left[(\mathbf{d}/2) \times (q\mathbf{E})\right] + \left[(-\mathbf{d}/2) \times (-q\mathbf{E})\right] = q\mathbf{d} \times \mathbf{E}.\end{aligned}$$ thus a dipole **p**=q**d** in a uniform field **E** experiences a torque: $$\mathbf{N} = \mathbf{p} \times \mathbf{E}.$$ 
force on polar molecules	if the field is nonuniform, F_+ doesn't exactly balance out F_- > there will be a torque and a force on the dipole > the force on a dipole in a nonuniform field is: $$\mathbf{F} = \mathbf{F}_+ + \mathbf{F}_- = q(\mathbf{E}_+ - \mathbf{E}_-) = q(\Delta \mathbf{E}),$$ assume the dipole is very short > we can approximate a small change in E_x as: $$\Delta E_x \equiv (\nabla E_x) \cdot \mathbf{d},$$ thus in 3D: $$\Delta \mathbf{E} = (\mathbf{d} \cdot \nabla)\mathbf{E},$$ therefore: $$\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}.$$

4.1.4 polarization

polarization **P**	place a material in a uniform field > if the material is neutral, there will be dipoles induced in the material the material consists of dipoles, they will experience a torque, aligning them > either way, there will be a lot of dipoles pointing along the direction of the field > the material becomes polarized, where we define the polarization **P** as: $$\mathbf{P} \equiv \textit{dipole moment per unit volume},$$

4.2 the field of a polarized object

4.2.1 bound charges

potential of a polarized object

Consider a polarized object, consisting of aligned dipoles
> the field of one dipole is given by:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\frac{\mathbf{p}\cdot\hat{\boldsymbol{\imath}}}{\imath^2}.$$

now, we have a dipole moment $\mathbf{p} = \mathbf{P}\,d\tau'$ in each volume element dτ'
> the total potential is:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int_{\mathcal{V}}\frac{\mathbf{P}(\mathbf{r}')\cdot\hat{\boldsymbol{\imath}}}{\imath^2}\,d\tau'.$$

now we know that:

$$\nabla'\left(\frac{1}{\imath}\right) = \frac{\hat{\boldsymbol{\imath}}}{\imath^2},$$

so we have:

$$V = \frac{1}{4\pi\epsilon_0}\int_{\mathcal{V}}\mathbf{P}\cdot\nabla'\left(\frac{1}{\imath}\right)d\tau'.$$

which we can simplify using (5) on the front cover:

$$V = \frac{1}{4\pi\epsilon_0}\left[\int_{\mathcal{V}}\nabla'\cdot\left(\frac{\mathbf{P}}{\imath}\right)d\tau' - \int_{\mathcal{V}}\frac{1}{\imath}(\nabla'\cdot\mathbf{P})\,d\tau'\right]$$

Stokes theorem tells us:

$$V = \frac{1}{4\pi\epsilon_0}\oint_{\mathcal{S}}\frac{1}{\imath}\mathbf{P}\cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0}\int_{\mathcal{V}}\frac{1}{\imath}(\nabla'\cdot\mathbf{P})\,d\tau'.$$

the first term looks like the potential of a surface charge with density defined as:

$$\sigma_b \equiv \mathbf{P}\cdot\hat{\mathbf{n}}$$

with **n** the normal unit vector
the second term looks like the potential of a volume charge:

$$\rho_b \equiv -\nabla\cdot\mathbf{P}.$$

then we have:

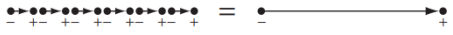
$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\oint_{\mathcal{S}}\frac{\sigma_b}{\imath}\,da' + \frac{1}{4\pi\epsilon_0}\int_{\mathcal{V}}\frac{\rho_b}{\imath}\,d\tau'.$$

>> the potential of a polarized object is the same as that produced by a charge with:

$$\sigma_b \equiv \mathbf{P}\cdot\hat{\mathbf{n}} \quad \text{and} \quad \rho_b \equiv -\nabla\cdot\mathbf{P}.$$

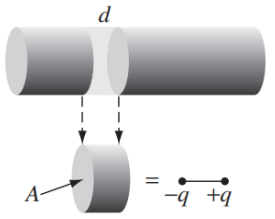
4.2.2 physical interpretation of bound charges

interpretation of bound charges

imagine a long string of dipoles 
> the head of one cancels out the tail of the outer
> however, at the ends of the string this doesn't happen
> it's as if we took an e^- form the head, and moved it to the tail of the string

>> the net charge is the *bound* charge, because it cannot be removed
> in a dielectric, every e^- is bound to an atom

bound charge mathematically





to calculate the amount of bound charge, examine the 'tube' of dielectric parallel to **P**
> the dipole moment of a tiny chunk is PAd, with A the cross-section area
> we can rewrite this as qd
> the bound charge that piles up at the righthand of the tube is therefore:

$$q = PA.$$

if the slices are sliced off perpendicularly, the surface charge density is:

$$\sigma_b = \frac{q}{A} = P.$$

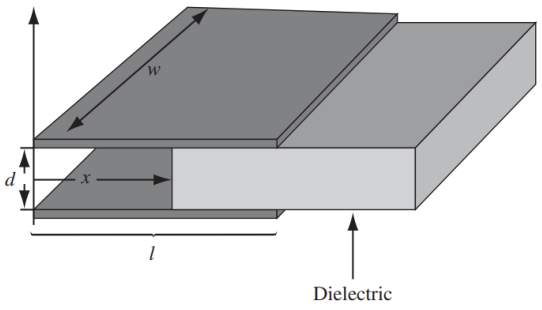
however, if it is sliced of at an angle θ:

$$\sigma_b = \frac{q}{A_{\text{end}}} = P\cos\theta = \mathbf{P}\cdot\hat{\mathbf{n}}.$$

>> the effect of polarization is to paint a bound charge σ=**Pn** over the surface

nonuniform polarization	we get accumulations of bound charge within the material, as well as on the surface > a diverging **P** results in a pileup of negative charge > we have: $\int_{\mathcal{V}} \rho_b \, d\tau = -\oint_{\mathcal{S}} \mathbf{P} \cdot d\mathbf{a} = -\int_{\mathcal{V}} (\nabla \cdot \mathbf{P}) \, d\tau.$ this is true for any volume V, thus we indeed find again: $\rho_b = -\nabla \cdot \mathbf{P},$
4.3 the electric displacement	
4.3.1 Gauss's law in the presence of dielectrics	
electric displacement	polarization causes accumulation of bound charge in the dielectric with: $\rho_b = -\nabla \cdot \mathbf{P}$ and on the surface: $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ now we can consider the free charge of the field > might consist of e^- or ions or anything that moves freely > *free charge is charge that is not associated with* ***P*** > the total charge of a dielectric is then: $\rho = \rho_b + \rho_f,$ then Gauss's law reads, for the total field **E**: $\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f,$ we can combine these terms: $\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f.$ define the *electric displacement*: $\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P},$ then: $\nabla \cdot \mathbf{D} = \rho_f,$ or integral $\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}},$
4.3.2 a deceptive parallel	
parallel between E and D	for D there exists no 'Coulomb's law' $\mathbf{D}(\mathbf{r}) \neq \frac{1}{4\pi} \int \frac{\hat{\boldsymbol{\imath}}}{\imath^2} \rho_f(\mathbf{r}') \, d\tau'.$ and the curl of D isn't zero: $\nabla \times \mathbf{D} = \epsilon_0 (\nabla \times \mathbf{E}) + (\nabla \times \mathbf{P}) = \nabla \times \mathbf{P},$
4.3.3 boundary conditions	
boundary conditions	we need for the perpendicular components: $D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} = \sigma_f,$ and the discontinuity parallel needs to be: $\mathbf{D}^{\parallel}_{\text{above}} - \mathbf{D}^{\parallel}_{\text{below}} = \mathbf{P}^{\parallel}_{\text{above}} - \mathbf{P}^{\parallel}_{\text{below}}.$ in the presence of an E we can rewrite $E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}} = \frac{1}{\epsilon_0}\sigma, \qquad \mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = \mathbf{0}.$

4.4 linear dielectrics	
4.4.1 susceptibility, permittivity, dielectric constant	
electric susceptibility χ_e	the polarization P due to an applied electric field E is: $$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}.$$ >> materials which obey this eq are *linear materials*
permittivity ε	in linear media we have: $$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 (1 + \chi_e) \mathbf{E},$$ thus **D** proportional to **E**: $$\mathbf{D} = \epsilon \mathbf{E},$$ where we define the permittivity: $$\epsilon \equiv \epsilon_0 (1 + \chi_e).$$
relative permittivity / dielectric constant ε_r	define: $\epsilon_r \equiv 1 + \chi_e = \frac{\epsilon}{\epsilon_0}$ which is depended on the type of material
curls of **P** and **D**	now that **P** and **D** are proportional to **E**, do their curls vanish like **E** does? > no: for the boundary of the material, there is a different $\varepsilon_0\chi_e$ on both sides > around this loop the curl doesn't vanish however, if the space is entirely filled with a homogeneous linear dielectric, then: $$\nabla \cdot \mathbf{D} = \rho_f \quad \text{and} \quad \nabla \times \mathbf{D} = \mathbf{0},$$ thus, **D** can be found from the free charge as though the dielectric were not there: $$\mathbf{D} = \epsilon_0 \mathbf{E}_{\text{vac}},$$ with $\mathbf{E}_{\text{vac}}$ the field the same free charge distribution would produce in a vacuum > we therefore have: $$\mathbf{E} = \frac{1}{\epsilon}\mathbf{D} = \frac{1}{\epsilon_r}\mathbf{E}_{\text{vac}}.$$
susceptibility tensor	in a crystal its easier to polarize in one direction as supposed to another, thus: $$\left.\begin{aligned} P_x &= \epsilon_0(\chi_{e_{xx}} E_x + \chi_{e_{xy}} E_y + \chi_{e_{xz}} E_z) \\ P_y &= \epsilon_0(\chi_{e_{yx}} E_x + \chi_{e_{yy}} E_y + \chi_{e_{yz}} E_z) \\ P_z &= \epsilon_0(\chi_{e_{zx}} E_x + \chi_{e_{zy}} E_y + \chi_{e_{zz}} E_z) \end{aligned}\right\}$$ which is a tensor
4.4.2 boundary value problems with linear dielectrics	
boundary conditions	in linear dielectric the bound charge density ρ_b is proportional to free charge density ρ_f: $$\rho_b = -\nabla \cdot \mathbf{P} = -\nabla \cdot \left(\epsilon_0 \frac{\chi_e}{\epsilon} \mathbf{D}\right) = -\left(\frac{\chi_e}{1 + \chi_e}\right) \rho_f.$$ within such a dielectric the potential obeys Laplace's equation > the boundary conditions are: $$\epsilon_{\text{above}} E^{\perp}_{\text{above}} - \epsilon_{\text{below}} E^{\perp}_{\text{below}} = \sigma_f,$$ or thus: $$\epsilon_{\text{above}} \frac{\partial V_{\text{above}}}{\partial n} - \epsilon_{\text{below}} \frac{\partial V_{\text{below}}}{\partial n} = -\sigma_f,$$ with the potential continuous: $$V_{\text{above}} = V_{\text{below}}.$$
4.4.3 energy in dielectric systems	
work on capacitor	to charge op a capacitor it takes a work: $$W = \tfrac{1}{2} C V^2.$$ its capacitance is given by: $$C = \epsilon_r C_{\text{vac}},$$

work on a capacitor	consider a dielectric-filled capacitor fixed in position > we bring in free charge a bit at a time > as ρ_f increases by amount $\Delta\rho_f$, the polarization changes > the work done on the incremental free charge is: $$\Delta W = \int (\Delta\rho_f) V \, d\tau.$$ since $\nabla\cdot D=\rho_f$ we have $\Delta\rho_f=\nabla\cdot(\Delta D)$, thus: $$\Delta W = \int [\nabla \cdot (\Delta \mathbf{D})] V \, d\tau.$$ with: $$\nabla \cdot [(\Delta \mathbf{D})V] = [\nabla \cdot (\Delta \mathbf{D})]V + \Delta \mathbf{D} \cdot (\nabla V),$$ and hence, via integrating by parts: $$\Delta W = \int \nabla \cdot [(\Delta \mathbf{D})V] \, d\tau + \int (\Delta \mathbf{D}) \cdot \mathbf{E} \, d\tau.$$ the divergence theorem terns the first term into a surface integral > this vanishes if we consider all space > the work done is: $$\Delta W = \int (\Delta \mathbf{D}) \cdot \mathbf{E} \, d\tau.$$ if the material is a linear dielectric, we have $D=\varepsilon E$, thus: $$\tfrac{1}{2}\Delta(\mathbf{D}\cdot\mathbf{E}) = \tfrac{1}{2}\Delta(\epsilon E^2) = \epsilon(\Delta\mathbf{E})\cdot\mathbf{E} = (\Delta\mathbf{D})\cdot\mathbf{E}$$ giving us: $$\Delta W = \Delta\left(\frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\, d\tau\right).$$ then the total work done is: $$W = \frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\, d\tau,$$
difference between formulas	In chapter 2 we derived the formula: $$W = \frac{\epsilon_0}{2}\int E^2 \, d\tau.$$ however, what is the difference between this one and the one we just derived? 1: bring in all the charges one by one with tweezers and glue each on down in position > if we want to consider the work done via this type of assembly, use: $$W = \frac{\epsilon_0}{2}\int E^2 \, d\tau.$$ >> this does NOT consider the stretching of dielectric molecules 2: bring in free charges one by one, allowing the dielectric to responds as it sees fit > the work done via this type of assembly is given by: $$W = \frac{1}{2}\int \mathbf{D}\cdot\mathbf{E}\, d\tau,$$ >> this accounts for the "spring" energy of the dielectric molecules
4.4.4 forces on dielectrics	
fringing field of a capacitor	We always pretended a capacitor doesn't have a E field outside its plates > however, there exists a fringing field around the edges > for if the field terminated abruptly, the line integral of E around the edge != 0 consider a dielectric that's half inserted into the capacitor > the dielectric will develop a charge on its surface > originally, the dielectric wouldn't be pulled inside the capacitor > however, the fringing field has a net force pushing the dielectric inside 

work due to fringing fields	consider the same setup as before > let F_{me} be the force I exert to counteract the electrical force F on the dielectric, then: $$dW = F_{\text{me}}\, dx,$$ since F_{me}=-F, we have: $$F = -\frac{dW}{dx}.$$ the energy stored in the capacitor is: $$W = \tfrac{1}{2} C V^2,$$ with: $$C = \frac{\epsilon_0 w}{d}(\epsilon_r l - \chi_e x),$$ assume the total charge on the plates is held constant as the dielectric moves > in terms of Q=CV, we have $$W = \frac{1}{2}\frac{Q^2}{C},$$ then the force: $$F = -\frac{dW}{dx} = \frac{1}{2}\frac{Q^2}{C^2}\frac{dC}{dx} = \frac{1}{2}V^2\frac{dC}{dx}.$$ but: $$\frac{dC}{dx} = -\frac{\epsilon_0 \chi_e w}{d},$$ thus: $$F = -\frac{\epsilon_0 \chi_e w}{2d} V^2.$$